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Hybrid ensemble approaches for cardiovascular disease prediction: Leveraging interpretable AI for clinical insight

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ABSTRACT

Objectives: Cardiovascular diseases are one of the most common and serious conditions that affect people's health. The use of early-stage signs to diagnose cardiovascular disease is a major problem in the modern world. However, there is a need for less expensive tools that detect cardiovascular disease without overbearing procedures.

Methods: In this study, we develop ensemble hybrid and blending techniques for the early detection of cardiovascular disease. Firstly, the proximity weighted random affine shadow sampling is utilized to reduce risk of bias model learning. Next, principal component analysis extracts most relevant features to improve accuracy and computational time. After that, we developed two novel ensemble models: hybrid and blending. A hybrid model combines convolutional neural networks and temporal convolutional networks, while blending model places convolutional neural networks and temporal convolutional networks, and a deep belief network on base layer and highway networks on meta layer to enhance cardiovascular disease detection performance. A combination of 10-fold cross-validation on clinical cardiovascular data showed both ensemble methods surpassing conventional models achieving peak results through the blending approach.

Results: The blending model reaches 0.914 accuracy and 0.9097 F1-score yet the hybrid model shows 0.8496 accuracy with 0.8496 F1-score. The project utilizes Local Interpretable Model-Agnostic Explanations (LIME) and Shapley Additive Explanations (SHAP) through explainable artificial intelligence (XAI) techniques for achieving dependability and transparency. The methods deliver explanations to clinicians by showing how clinical characteristics influence the predicted results.

Conclusion: The proposed ensemble model demonstrates strong predictive capabilities combined with interpretability which allows its use in real-time operations for cardiovascular disease prediction.

1. Introduction

Cardiovascular diseases (CVD) are defined as a group of conditions affecting blood that lead to heart disease. They include congenital conditions, hypertension, coronary artery disease, and rhythm disturbances. According to the World Health Organization, they continue to take the lives of 18 million people globally, or one in eight deaths, yearly. In fact, 85 percent of these deaths are either from heart attacks or strokes [1]. This can be increased by aging populations in both developed and developing countries, and it can be linked to several factors, including a sedentary lifestyle, poor eating habits, a lack of movement, and increasing occupational stress. The risk factors are genetic predisposition, high arterial hypertension, and elevated blood lipids. Early detection and treatment are crucial because atrial fibrillation, the most common arrhythmia diagnosed in a clinical environment, is an independent risk factor for ischemic stroke.

Early diagnosis of CVD is critical to reduce mortality and to improve patient health outcomes. A hallmark of many current diagnostic practices is reliance on the visible cycle of the heart rate or rhythm that is recorded, such as in Electrocardiograms (ECGs). However, identifying arrhythmic events remains difficult because of their irregular and unpredictable nature. Following all the technologies that came before it, Deep Learning (DL) has revolutionized the way that big, multidimensional datasets are analyzed. It has been shown that DL models can increase diagnostic accuracy by picking up on subtle patterns in complicated signals, such as cardiac beats. Consider Deep Neural

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Network (DNN) models, which have shown impressive results in detecting erratic heartbeats. Their sensitivity ratings are around twice as high as those of ordinary cardiologists, and they also reduce the probability of misclassification.

For doctors to diagnose a variety of cardiac conditions, such as arrhythmias and acute coronary syndromes, and to evaluate the state of the heart, ECGs are essential. Over the past few years, Machine Learning (ML) and DL have advanced to such an extent that CVD prediction and management have been greatly improved. The study shows that DL models currently outperform traditional ML methods and demonstrate increased resilience while managing diverse and high-dimensional healthcare data. For instance, Yonder et al. in Ref. [2] reported that a DNN model can achieve an accuracy of 98.15 % and a sensitivity of 98.67 % in coronary heart disease prediction, higher than traditional models with Support Vector Machine (SVM).

However, redundancies, incompleteness, and data sensitivity limit their efficacy. Healthcare companies worldwide employ Electronic Medical Records (EMR), Electronic Health Records (EHR), diagnostic imaging, and medical device data to investigate their potential. The unstructured, dimensional medical dataset is inherently complex and necessitates novel computational methods for accurate diagnosis and prognosis. Furthermore, the dataset is frequently imbalanced, with unequal class distributions that are highly predictive of class marks.

This study aims to address these challenges through the following contributions:

- **Data Balancing:** The Proximity Weighted Random Affine Shadow sampling (ProWRAS) balancing technique is employed to mitigate the problem of highly imbalanced data to reduce model learning.
- **Feature Extraction:** The Principal Component Analysis (PCA) is used to extract important features to achieve enhanced accuracy with reduced computational time.
- **Models Development:** The two new ensembles, including a hybrid and a blending model, are proposed to improve the accuracy of the CVD prediction. A hybrid ensemble comprises Convolutional Neural Network (CNN) and Temporal Convolutional Network (TCN). In contrast, the blending ensemble places CNN, TCN, and Deep Belief Network (DBN) on the base layer and Highway Network (HN) on the meta layer.
- **Explainable Artificial Intelligence Utilization:** Local Interpretable Model-agnostic Explanations (LIME) and Shapley Additive Explanation (SHAP) are used to deep-dive in both hybrid and blending ensemble predictions.
- **Performance Evaluation:** The performance of the proposed model is then compared with the most recent techniques in terms of accuracy, sensitivity, and precision.

1.1. Statement of significance

1.1.1. Problem or issue

Cardiovascular disease (CVD) continues to be a leading cause of death worldwide. Existing diagnostic methods often fail to accurately predict the risk of CVD due to the complexity and variability in patient data. The lack of transparency in many predictive models further limits their clinical utility.

1.1.2. What is already known

Machine learning and deep learning models have shown promise in predicting CVD risk. However, many of these models act as "black boxes," making it difficult for clinicians to interpret their decisions. While techniques like LIME and SHAP have been used to explain model predictions, they have not been widely integrated into ensemble approaches for CVD prediction.

1.1.3. What this paper adds

This paper presents a novel ensemble deep learning approach for CVD prediction that combines multiple models, including CNN, TCN, and DBN, with a Highway Network meta-model. The proposed method not only improves prediction accuracy but also enhances model transparency through LIME and SHAP, providing clear, interpretable explanations of the role of key features like diabetes, smoking history, and exercise in predicting CVD risk. Additionally, the model is compared with established risk scores like the Framingham Risk Score and evaluated using 10-fold cross-validation, demonstrating superior performance in terms of accuracy and reliability. This study bring up (i) A Hybrid (CNN + TCN) and a Blending (CNN + TCN + DBN-HN) architecture for CVD screening on survey-style risk factors; (ii) a class-imbalance strategy (ProWRAS) with fold-wise PCA to reduce leakage; (iii) thorough evaluation with PR-AUC/ROC-AUC and confusion-matrix analysis, plus XAI to surface clinician-relevant drivers.

2. Related work

In the recent few years, numerous deep learning techniques have been proposed by the research community for the evaluation, early diagnosis, and right treatment. In this section, there have been some studies using DL models that were conducted to help the healthcare sectors for better treatment and disease prediction and to help in decision-making. Recent research has employed various ML and DL techniques to forecast and predict diseases in healthcare applications. Below, we categorize the existing work based on the methods employed and the application domains targeted. Additionally, an analysis of the limitations of existing approaches is presented to highlight the innovation of the current research.

2.1. Machine learning approaches

Using the above-mentioned methodologies can lead to several major issues, including the requirement for features engineering, large resource utilization, substantial data preprocessing, and reliable classification findings. In Ref. [3], Waqar et al. suggested that a research project offers a practical and affordable way to forecast heart attacks reliably and accurately. Without using feature extraction, it makes use of a University of California Irvine (UCI) dataset and a variety of ML methods to predict heart attacks. Additionally, there is an uneven distribution of both positive and negative categories in the provided dataset, which hinders performance. To address the imbalanced data, the suggested study employs a Synthetic Minority Oversampling Technique (SMOTE). The proposed approach eliminated the need for technical features to categorize the dataset. Given that feature engineering is frequently an expensive procedure, this has resulted in an effective solution. The findings demonstrate that when appropriately calibrated, SMOTE based Artificial Neural Networks (ANNs) outperformed several current systems and all different models among all ML techniques.

In [4], the author introduces MedAi, a smartwatch-based forecasting model for many diseases like myocardial infarction, diabetes, hypertension, and many more through ML techniques. The dataset comprises patient bodily statistics that can be arranged from healthcare centers through consent. The authors used many ML algorithms to make inquiries about the best technique. Results reveal that the Random Forest (RF) technique has the best performance in predicting the above-mentioned diseases with an accuracy of 99.4 %. In Ref. [5], Behera et al. suggested a hybrid model created by merging the SVM and the updated Particle Swarm Optimization (PSO) technique. The collection of datasets comes from the UCI ML library. All results are compared against the standards of error, F1 score, recall, and classification accuracy. The CPSOSVM, SVM, and hybrid are compared to the results attained.

In [6], Li et al. suggested an ML-based system for the treatment of heart disease that is both successful and accurate. The algorithm was

created by reducing unused and superfluous features using industry-standard feature selection methods. SVM, Logistic Regression (LR), ANN, and Decision Tree (DT) are some of the classification techniques used by the system to address the issue of feature selection. In Ref. [7], feature selection approaches are used to increase classification accuracy and reduce processing time. Furthermore, tuning hyper-parameters and learning the best practices for model evaluation have both been accomplished using the leave-one-subject-out cross--validation method. The production measuring parameters are used to assess the classifiers' capabilities. The efficacy of the classifiers has been assessed on the characteristics picked by feature selection techniques. The results of the experiments show that the proposed feature selection technique is suitable for establishing a high-level intelligent network to determine heart disease using a classifier SVM.

In [8], Hemalatha et al. proposed research to segment and categorize blood cells using the K-means technique and with the aid of an imaging cleaning approach. Previously, a hem cytometer, other instruments, and certain chemicals were used to manually calculate blood cells. This method is both overdue and challenging.

2.2. Deep learning approaches

In [9], Zhou et al. talks about the early warning and forecasting technique for heart failure with the help of DL. This suggested model includes a risk feature selection technique to recognize the main risk factors that hold the most efficient and valuable data for forecasting. The suggested model train on the real project can gain an accuracy of 98.5 %, which is more impressive than any previous work. In Ref. [10], Liza et al. inquire about the eight ML techniques, such as DT, Adaptive Boosting (AB), Gaussian Naive Bayes (GNB), Xtreme Gradient Boosting (XGB), RF, SVM, and K-Nearest Neighbors (KNN). A well-known ML database, the Wisconsin dataset, was located at UCI. The presentation of the work is evaluated in connection to recall, ROC score accuracy, F1-score, and precision. With an accuracy of 99.20 % and a Receiver Operating Characteristics Area Under the Curve (ROC-AUC) score of 99 %, AB and RF perform the best out of all of them.

In DL, a technique based on Synthetic Neural Networks (SNN) has recently come into prominence as a powerful ML tool with the potential to fundamentally alter the area of Artificial Intelligence (AI) [11]. Recent developments in parallelization, significant data storage, and computational power, in addition to the technology's capacity to automatically optimize features and semantically deduce from the incoming data, have all contributed to its rapid acceptance. In Ref. [12], Ravi et al. provide a censorious evaluation of the comparative merit, potential drawbacks, and prospects of the DL algorithms while providing a comprehensive, up-to-date review of research using the method in health informatics. The research mainly concentrates on significant DL applications in pervasive sensing, healthcare information systems, medical imaging, and health services.

In [13], Zhang et al. highlighted how the Internet of Things (IoT)--based healthcare system has shown to be the most effective in treating CVD in medical environments. On synthetic and practical functional datasets, an immense number of tests were run. The results appear to be that the suggested technique can get a more precise and physically consistent evaluation. Also, the suggested method is more related to and deeply modifies IoT and DL in smart healthcare. In Ref. [14], Banna et al. people use social media as a means of showing their emotions of anxiety, fear, and depression due to fewer outdoor activities. By focusing on depression, the authors will analyze the effect of the Coronavirus Disease (COVID-19) on mental health through the Twitter microblog. The authors suggest a novel pipeline, which can identify depressive microblogs with an accuracy of 99.42 %, worked on Recurrent Neural Network (RNN) and CNN. The goal was to find the depressive tweets by preprocessing using multiple natural language techniques.

In [15], the author presented a framework for ensemble learning using several neural network models together with a technique to aggregate random undersampling. The suggested approach involves selecting characteristics as part of a data preprocessing stage to enhance the performance of the classification algorithms. A group of classifiers with a Bidirectional Long Short-Term Memory (BiLSTM) or Bidirectional Gated Recurrent Unit (BiGRU) algorithm based on a CNN model had an excellent performance in classification with accuracy and an F1-score within 91 % and 96 % for the various types of CVD, according to experiments with unidirectional and bidirectional neural network models. Arumugham et al. in Ref. [16] proposed the early prediction of Chronic Kidney Disease (CKD) through a DNN with an adaptive moment estimation optimization function. The healthcare application needs to be accountable for the forecasting of the black box algorithm to create opinions about the model forecasting. Thus, the forecasting of the DNN-CKD model is elucidated by LIME. This data of diagnostic patients is trained on a 5-layered DNN with three hidden layers. The DNN-CKD model produces a precision of 98.75 % and an ROC-AUC score of 98.86 % over unseen data in the detection of CKD prevalence.

In [17], Ma et al. developed DL techniques for gait detection for Parkinson's Disease (PD) patients. This work advances an accountable learning architecture that requires DL, feature evaluation, data balancing mechanism, data selection, and ML. Different algorithms were used to analyze the outcomes from both patients and healthy individuals. The research planning applications that are evolving in this work notably increase the comprehensive rate of recognition, mainly when we used the suggested architecture with XGB and DL to attain precision rates as high as 97.32 % and 98.41 %.

In [18], Sarveshwaran et al. stated that the healthcare industry uses DNN to make better care delivery, which is cost-effective and time-saving. A DNN is widely known for its diagnostic approach. Despite that, DNN is also being used for healthcare management decisions. In Ref. [19], the author suggested that DNN is also used for the classification of real-life disease issues. By decreasing the training time, carrying out swift classification, and upgrading the parameters, the performance of DNN will be improved. Strassen's rectilinear fine tube bouncing training algorithm is a new hybrid model that we proposed by fusing Strassen's theorem and the bouncing training technique.

In [20], Musthafa et al., using statistical information first, enhances Snake Swarm Optimization (SSO) with the bat technique for lung nodule segmentation. To decrease the dimensionality issue, we illustrate a chaotic Atom Search Optimization (ASO) algorithm to choose the finest characteristic between multiple features. For nodule forecasting and grading, we created a hybrid learning-based DNN classifier.

2.3. Ensemble approaches

The authors in Ref. [21] suggested creating a hybrid deep learning approach in this study. RNN and LSTM hybrid techniques have been used to classify synthetic data using various cross-validations. RNN uses three distinct functions of activation in the classification procedure. A few preprocessing techniques were applied to the information to classify and arrange it equitably. Three methods have been used to extract features: interpersonal, bigram, and density based. The authors evaluated system performance during the experiment using a range of ML and DL techniques. The results section displays the classification accuracy of each algorithm. The author of [18] presented that we merge a CNN with the LSTM to attain higher accuracy than conventional ML techniques. As a result, a hybrid model of CNN and LSTM techniques was used to distinguish between normal and abnormal. This model gives an accuracy of 89 %, and it was authenticated by using k-fold cross-validation techniques. The proposed model provided more accurate results and more acceptable performance than existing ML algorithms.

In [22], Sharmin et al. suggested a hybrid reliable technique for breast cancer detection that merges the ability of DL to employ a pre-trained ResNet50V2 technique and grouped based ML techniques. Using a build-up of samples of various sizes from freely accessible breast histopathology image-based Invasive Ductal Carcinoma (IDC) datasets,

we conducted extensive studies. We get a considerable accuracy rate of 95 %. We also conclude that the Light Gradient Boosting (LGB) is the most satisfactory ML technique with ResNet50V2 architecture. The results of this study provide a remarkable endowment to the prediction of breast cancer. In this paper [23], Krishnan et al. present an innovative DL hybrid model that combines LSTM, Adam optimizer, and multiple GRU in an RNN to predict cardiac disease. With an astounding accuracy of 98.6876 %, the suggested model outperformed the current RNN model. RNN-based models that are now in use have achieved an accuracy of 98.23 %, whereas DNNs have achieved 98.5 %. Studies with numerous customized models revealed that the optimal model performs best when multiple GRU and RNN are combined with SMOTE. This is the highest accuracy result for an RNN using Cleveland datasets, and it has a lot of potential for early heart disease prediction in patients.

2.4. Limitations in existing approaches

While the works demonstrate promising results, several limitations persist. Many models, such as those in Refs. [15,23], suffer from high computational costs and long training times, which hinder their practical application. Additionally, methods like SMOTE, used to address class imbalance in Ref. [3], face challenges in the presence of outliers and overfitting. Moreover, recurrent models like RNN and LSTM often struggle with the vanishing gradient problem, as noted in Ref. [21], and fail to capture long-term dependencies effectively. Similarly, hybrid models like those in Ref. [23] show improvements but at the cost of increased susceptibility to overfitting.

For this reason, there is a demand for suitable model architectures that trade off performance, efficiency, and interpretability. Furthermore, addressing issues related to overfitting, feature engineering, and computational cost remains crucial to advancing the state-of-the-art in healthcare predictions.

2.5. Innovation of this research

The limitations mentioned above are subject to be overcome by the current research, which proposes a more efficient and robust ensemble model architecture for CVD prediction. Finally, the approach balances data through less sensitive data balancing techniques for outliers and ensemble DL models without the vanishing gradient problem carrying for dependency in long term. The research also discusses ways to enhance computational efficiency without compromising the model accuracy and how to relieve the high computational cost of traditional methods. This study makes a unique contribution by presenting models that not only improve prediction accuracy but also ensure better generalization, interpretability, and resource efficiency, thus paving the way for more practical applications in healthcare.

3. Proposed model for cardiovascular disease prediction

In this section, we proposed our system model shown in Fig. 1. We provide an early CVD detection method based on ensemble hybrid and blending approaches. After obtaining our dataset, we preprocess it, use balancing strategies, and then run models for hybrid and blending. There are two models in hybrid and four models in our blending. The hybrid is created using CNN and TCN, and in blending base models are CNN, TCN, and DBN. HN is used as a meta-model and compared with all of the baseline models. Considering this, our hybrid and blending model is used for the early detection of CVD prediction. In this model, the pipeline includes preprocessing, data balancing, and advanced DL techniques that are used for CVD classification. The **CNN** backbone consists of stacked 1-D convolutional blocks (Conv1D → ReLU → BatchNorm → Dropout), followed by global pooling and a sigmoid classification head. The **TCN** backbone uses **causal, dilated convolutions** with residual/skip connections to capture long-range temporal dependencies; each block applies (Dilated Conv1D → ReLU →

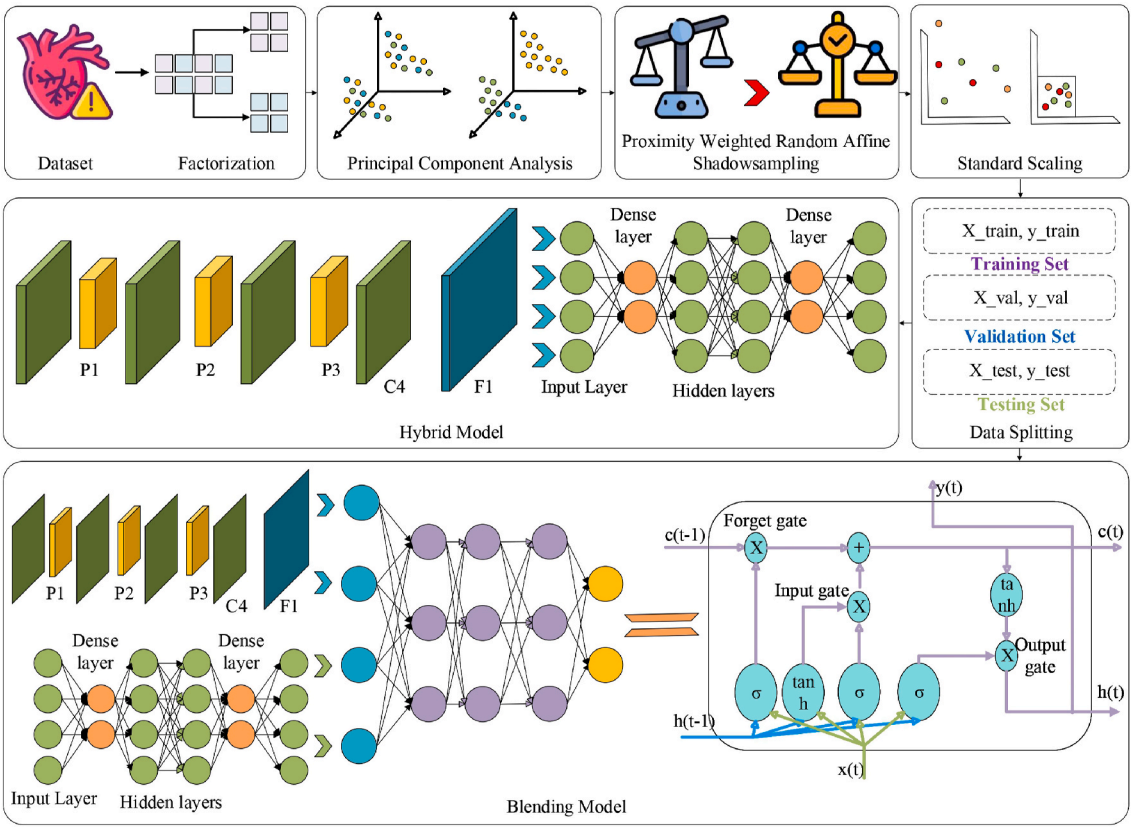


Fig. 1. Hybrid & blending ensemble for early CVD prediction.

normalization → Dropout) with increasing dilations. The **DBN** is implemented as a **stack of Restricted Boltzmann Machines (RBMs)** trained layer-wise, then fine-tuned for classification. For the **blending ensemble**, logits (or calibrated probabilities) from **CNN, TCN, and DBN** serve as inputs to a **Highway Network (HN) meta-learner**, which combines transformed features. This setup lets the meta-learner retain salient base-model signals while adding non-linear corrections, improving discrimination with minimal overhead First, categorical data is preprocessed to integers for it to contribute to DL algorithms. In the Hybrid model, CNN and TCN are trained separately on the same preprocessed folds; their outputs are then concatenated and passed through a small classifier head for prediction. In the Blending model, CNN, TCN, and DBN are trained with cross-validation to produce out-of-fold predictions, which are combined into a meta-dataset. A Highway Network (HN) is then trained on this meta-dataset to learn how to weight the base model outputs. At inference, base predictions are passed to the trained HN, which produces the final decision. This avoids data leakage and makes the ensemble process explicit. Then, PCA is applied to lower dimensionality, keeping only the most important features. PCA was always fitted after class rebalancing (SMOTE/ProWRAS) on the training folds only, and the same transformation was then applied to the validation/test folds. This ensures that sampling operates in the full original feature space, while PCA provides a compact, decorrelated representation for downstream learners without information leakage. Prior to constructing the model, the dataset is normalized to standardize feature values. 80 % of the data are used for training, 20 % for testing for hybrid testing, and 60 % percent are used for training, 20 % percent are used for validation, and 20 % percent are used for testing for blending model analysis. For deep models (CNN, TCN, DBN), we fixed learning rate, batch size, and dropout values after light manual tuning guided by validation folds. For tree-based or linear baselines, we used standard defaults with limited grid search (e.g., number of estimators, maximum depth, or regularization strength). This pragmatic approach reduced computational cost while still ensuring stable performance; ablation toggles (PCA on/off, sampling methods, meta-learner variants) further demonstrate robustness across hyperparameter setting. The data-balancing mechanism of handling class imbalances is achieved with ProWRAS. As base models, two things are in the pipeline: first, a hybrid model, which is a combination of TCN and CNN, and second, a blending model that uses TCN, DBN, and CNN as base models and an HN as a meta model. Then, the two models combine the strengths of TCN and CNN in working with sequential and spatial data patterns, and the blending model combines predictions of various base models for improved overall performance. Finally, the processed data is classified to predict CVD accurately. For deep learning models, we used binary cross-entropy as the loss function and the Adam optimizer with an initial learning rate of 0.001. Early stopping was applied with a patience of 10 epochs based on validation loss to avoid overfitting. Experiments were conducted on an NVIDIA Tesla V100 GPU with 32 GB memory; classical ML baselines were run on CPU. These settings ensure reproducibility and provide a fair basis for comparing different model families.

3.1. Dataset

CVD, which is one of the most common chronic conditions in the United States of America (USA), affects millions of people each year and has an enormous financial effect on the financial condition of the country. CVD is the leading cause of mortality in the USA, accounting for around 647,000 fatalities per year. Chronic inflammation, elevated blood pressure, and hyperglycemia are all risk variables and causes of CVD. Although there are many different types of CVD, the most common ways that people with the condition discover they have it are through heart attacks, sudden cardiac arrest, or chest discomfort. This directs the attention of the healthcare facility to gathering disease data for its profit. The Centers for Disease Control (CDC) provided the annual Behavioral Risk Factor Surveillance System (BRFSS) data 2021, which was used in the data collection process for this study. The dataset was initially viewed on a local computer and comprised 438,693 records with 304 features in total. However, not all of the features were relevant to this specific study. As such, the author selected and included a certain subset of 19 features in the development of the DL models. As a result, the data that we receive has nearly 19 features in total. Because of this careful curation, there were fewer records overall; 308,854 data instances were used for analysis and model building.

3.2. Data-driven approach for cardiovascular disease (CVD) diagnosis

The disease CVD stands as the main reason behind global yearly deaths particularly in the United States with annual deaths reaching approximately 647,000. Increased effectiveness in CVD prediction and diagnosis methods stands essential due to its extensive influence on public health and economic stability. This study relies on 438,693 records and 304 features taken from the Centers for Disease Control (CDC) Behavioral Risk Factor Surveillance System (BRFSS) data of 2021. The analysis proceeded using 308,854 instances following the reduction of features to 19. Researchers have developed a specific dataset to enhance deep learning (DL) model performance for CVD prediction in minimal time.

3.3. Correlation heatmap: data insights and predictions

The correlation heatmap below highlights relationships among the various features relevant to CVD prediction. A heatmap provides a useful way to show the relationships between general health and exercise habits as well as smoking history and age category with diabetes and hypertension as well as BMI. The analyzed data demonstrated that overall health status presented meaningful positive links with the existence of diabetes and cardiovascular disorders. Those with bad general health tend to display increased possibilities of obtaining heart disease and related cardiovascular conditions. Doctors gain fundamental insights about CVD risk relationships through visual representation in the heatmap which shows the connections between different factors.

3.4. Impact on diagnosis and healthcare efficiency

The research adopts deep learning techniques together with visualization through correlation heatmaps to reduce medical expenses and examination times needed for CVD diagnosis. The current healthcare diagnostic methods take many hours while using numerous examinations with high expense costs. The proposed technique uses machine learning solutions to lower diagnosis costs through fast identification of patients with high risk factors represented by their health data. This model allows healthcare providers to better understand the reasoning behind their predictions. Medical staff can use heatmap visualization to recognize data relationships which enables them to make better care decisions for their patients. The diagnostic approach accelerates medical diagnosis while promoting accurate medical treatments that provide effective patient care at better costs.

3.5. Data preprocessing

To guarantee the accuracy and reliability of the data needed to train and assess neural network models, data preprocessing is an essential stage in the DL evaluation process. The data must be cleaned, transformed, and made ready for modeling using a variety of approaches.

3.5.1. Handling null values and duplicate records

The effectiveness of deep learning models can be greatly impacted by null values, sometimes referred to as missing data. Consequently, it's critical to recognize and manage null values appropriately. Feature engineering, deletion, and imputation are typical methods. Overfitting and bias can be introduced by duplicate values. Data integrity requires the

identification and removal of duplicate records. Since there were no null values or duplicate entries in our data. Thus, using data was virtually good.

3.5.2. Factorization and normalization

The technique of changing and modifying features to enhance the data description is called data factorization, as well as feature engineering. PCA, one-hot encoding, and normalizing are often used approaches in it. To standardize our data, we normalize it. It will thus receive equal training on the data throughout the training process and provide accurate predictions.

4. Discussion of results

In this section, a detailed analysis of our proposed models encompassing the hybrid and blending methods as well as their respective base models is presented. Furthermore, confusion matrices are used to compare the proposed model's performance. Additionally, explainable AI methods like LIME and SHAP are used to examine how the models make decisions. The outcomes show that the ensemble-based blending model achieves better classification performance with fewer misclassification errors than both the hybrid model and individual DL models.

4.1. Evaluation matrices

4.1.1. Accuracy

The accuracy score is an assessment statistic that calculates a model's proportion of accurate predictions compared to all predictions. It is computed by dividing the total number of predictions by the number of correct predictions:

$$\text{Accuracy} = TP + TN / TP + TN + FP + FN \tag{1}$$

4.1.2. Precision

One measure of the ML model's effectiveness is precision, or how well the model makes positive predictions. When we talk about precision, we're talking about the ratio of True Positives (TP) compared to all positive predictions:

$$\text{Precision} = TP / TP + FP \tag{2}$$

4.1.3. Recall

The ratio between the correctly predicted positive samples and actual TP samples is recall:

$$\text{Recall} = TP / TP + FN \tag{3}$$

4.1.4. F1-score

The F1-score is an evaluation statistic for ML that gives an idea about how accurate a model is. It combines model precision and recall rating. The this metric calculates the number of times a model will have correctly predicted during the whole dataset:

$$\text{F1-score} = 2 \times \text{Precision} \times \text{Recall} / \text{Precision} + \text{Recall} \tag{4}$$

4.1.5. Receiver operating characteristics area under the curve

ROC-AUC is used to assess a model's capacity to discriminate between positive and negative classes. It displays the True Positive Rate (TPR) at different classification thresholds plotted against the False Positive Rate (FPR). The ROC-AUC value falls between 0 and 1, with one denoting perfect classification and 0.5 denoting random guessing:

$$TPR = TP / TP + FN \tag{5}$$

$$FPR = FP / FP + TN \tag{6}$$

4.1.6. Precision-recall area under the curve

The PR-AUC assesses the precision vs recall trade-off at various classification thresholds. It is especially helpful for datasets that are imbalanced, meaning that there are more negative examples than positive ones. The PR-AUC shows how well the model can classify positive samples, while the PR curve plots precision against recall:

$$Precision = TP / TP + FP \tag{7}$$

$$Recall = TP / TP + FN \tag{8}$$

4.2. Results and discussion for proposed models

In this section, the performance of a hybrid model, proposed blending model and the base models are compared. Execution time, ROC-AUC, PRAUC, F1-score, accuracy, precision and recall are used to assess the models. The purpose is to demonstrate how well ensemble approaches to increase predictive accuracy and dependability. Fig. 2 shows a performance comparison of the proposed models, the blending model, hybrid model and the base models (CNN, TCN, DBN and HN), and Table 1 compares the F1, Accuracy and Recall of the proposed models.

The use of the four important evaluation metrics, namely, accuracy, precision, recall, and F1-score, is graphically compared between the models. The blending model achieves the best scores in the majority of the metrics with the highest scores. The results of its are 0.914 % accuracy, 0.961 % precision, 0.863 % recall, 0.909 % F1-score. Its superior performance lies in the ensemble nature of the blending model, which combines the predictions from the collective of the base models (i.e CNN, TCN, and DBN) through an HN as the meta model. The hybrid model with CNN and TCN combined achieves 0.872 %, 0.903 %, 0.836 %, and F1-score of 0.868 %. In the matter of predicting cardiovascular disease (CVD) uses KFCV (K-Fold Cross Validation) to evaluate the performance of the hybrid and Blending models. This method splits the dataset into 'K' subsets where each fold is used for training as well as a testing sample, hence it yields a robust estimate of the model performance across folds. Ablation-style results are provided in Supplementary Table 1, showing stepwise performance gains from baseline CNN through PCA, SMOTE, ProWRAS, hybridization, and ultimately blending strategies. The metrics employed here include accuracy, F1-score, recall, and precision, and these are a broad measure for ranking of the model's performance in terms of predictive power of CVD prediction with minimal false positives and negative outcomes. Although the ensemble models generally outperformed individual learners, in some folds the DBN alone achieved higher recall than the Hybrid configuration. This likely reflects DBN's ability to capture non-linear feature hierarchies with relatively fewer parameters, which can favor sensitivity. Such cases emphasize that ensembling does not guarantee universal gains across every metric, but rather provides more consistent overall performance.

The KFCV of the used hybrid model, predicts cardiovascular disease (CVD) is presented in Table 2. This includes accuracy, F1 score, recall and precision for several folds. Each fold's accuracy, F1-score, recall, and precision values are presented with an average of 0.8496 for all of them for the model's consistency in performance across varying data subsets. These results show that in predicting CVD, the hybrid model is very capable of, while achieving precision and recall balance. In addition to the core metrics, Supplementary Table 1 also reports agreement-based measures such as the Matthews Correlation Coefficient (MCC) and Cohen's κ. Both indicate strong consistency (MCC >0.80, $\kappa > 0.82$) for the proposed ensembles, underscoring their reliability on imbalanced data.

To compute the KFCV results for a Blending Model used in the prediction of cardiovascular disease (CVD) using Table 3. Key metrics to evaluate the performance of the model are Accuracy, F1-Score, Recall and Precision for various folds in this model. The results show that the Blending Model consistently performs well, with average accuracy of 0.914, F1-score of 0.9097, recall of 0.8633, and precision of 0.9616. These results reflect a strong balance between precision and recall, with

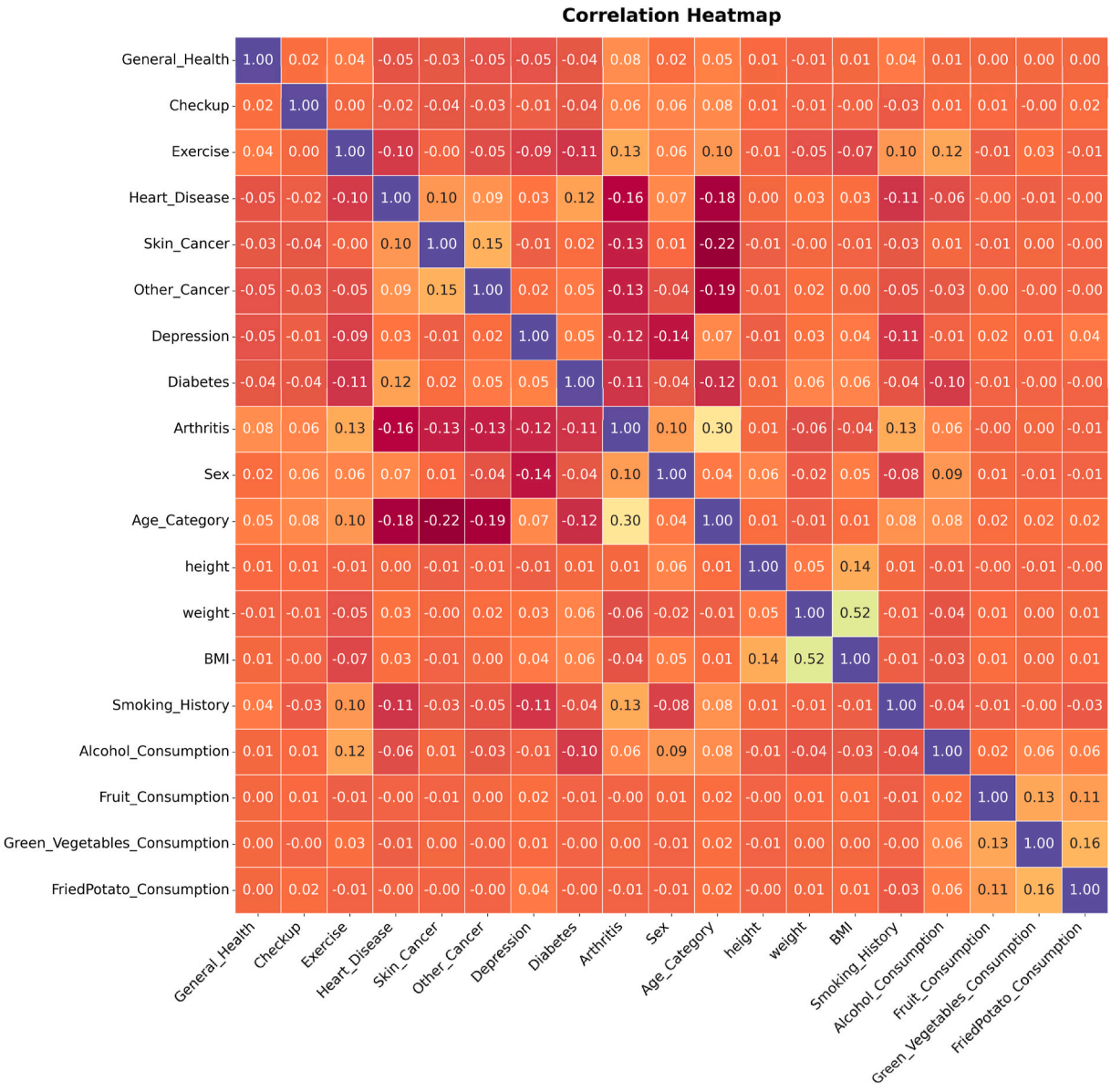


Fig. 2. Correlation heatmap of key features for cardiovascular disease prediction.

Table 1
Hybrid model, blending model, and base models results for CVD prediction.

Model	Accuracy	F1-Score	Recall	Precision	Execution Time (sec)
CNN	0.8206	0.8168	0.7961	0.8386	179
TCN	0.809	0.809	0.809	0.809	352
DBN	0.8051	0.8182	0.8726	0.7701	356
Highway	0.880	0.8827	0.8963	0.8695	194
Hybrid Model	0.8496	0.8496	0.8496	0.8496	212
Blending Model	0.914	0.9097	0.8633	0.9616	97

Table 2
Hybrid Model KFCV results for CVD Prediction.

Model	Accuracy	F1-Score	Recall	Precision
Hybrid Model	0.8874, 0.5000, 0.8767, 0.9010, 0.8897, 0.9029, 0.8944, 0.8699, 0.8950, 0.8793	0.8874, 0.5000, 0.8767, 0.9010, 0.8897, 0.9029, 0.8944, 0.8699, 0.8950, 0.8793	0.8874, 0.5000, 0.8767, 0.9010, 0.8897, 0.9029, 0.8944, 0.8699, 0.8950, 0.8793	0.8874, 0.5000, 0.8767, 0.9010, 0.8897, 0.9029, 0.8944, 0.8699, 0.8950, 0.8793
Average	0.8496	0.8496	0.8496	0.8496

particularly high precision values, indicating that the Blending Model is effective at identifying true positive cases of CVD while minimizing false positives. To quantify robustness, 95 % confidence intervals (mean ± 1.96·SD across folds) for accuracy, precision, recall, and F1 are included in Supplementary Table 1. These intervals confirm the stability of improvements across cross-validation folds, rather than being fold-specific artifacts.

Even though the hybrid model performs marginally worse than the blending model, it still outperforms the individual base models. The CNN has an F1-score of 0.816 % and achieves 0.820 % accuracy, 0.838 % precision, and 0.796 % recall among the base models. With an accuracy, precision, recall, and F1-score of 0.841 %, the TCN model exhibits comparable performance. The DBN model performs worse than

Table 3
Blending Model KFCV results for CVD Prediction.

Model	Accuracy	F1-Score	Recall	Precision
Blending Model	0.9194,	0.9171,	0.8879,	0.9483,
	0.9158,	0.9123,	0.8724,	0.9560,
	0.9129,	0.9079,	0.8556,	0.9671,
	0.9172,	0.9130,	0.8662,	0.9652,
	0.9141,	0.9103,	0.8688,	0.9561,
	0.9124,	0.9087,	0.8685,	0.9527,
	0.9126,	0.9082,	0.8613,	0.9605,
	0.9122,	0.9074,	0.8575,	0.9635,
	0.9136, 0.9101	0.9084,	0.8537,	0.9705,
		0.9037	0.8413	0.9761
Average	0.914	0.9097	0.8633	0.9616

the other models with an accuracy of 0.805 % and an F1-score of 0.818 %, suggesting that its probabilistic learning strategy is less successful. As the meta-model in the blending approach, the HN attains an F1-score of 0.882 %, 0.836 % accuracy, 0.880 % precision, and 0.896 % recall. The proposed model's resilience in producing accurate health predictions is demonstrated by this performance improvement.

Even though the hybrid model performs marginally worse than the blending model, it still outperforms the individual base models. The CNN has an F1-score of 0.816 % and achieves 0.820 % accuracy, 0.838 % precision, and 0.796 % recall among the base models. With an accuracy, precision, recall, and F1-score of 0.841 %, the TCN model exhibits comparable performance. The DBN model performs worse than the other models with an accuracy of 0.805 % and an F1-score of 0.818 %, suggesting that its probabilistic learning strategy is less successful. As the meta-model in the blending approach, the HN attains an F1-score of 0.882 %, 0.836 % accuracy, 0.880 % precision, and 0.896 % recall. The proposed model's resilience in producing accurate health predictions is demonstrated by this performance improvement. The proposed models, including the blending model, hybrid model, and base models (CNN, TCN, DBN, and HN), are shown in Fig. 3A, along with their execution times. The total time required by each model to finish the training and prediction process is represented by the execution time. Despite being an ensemble model, the blending model's efficiency is demonstrated by its fastest execution time of 97 s. Due to HN's lightweight design, which serves as the meta-model and integrates predictions from the base models, this performance advantage is explained. With a total execution time of 194 s, the CNN model is the second-fastest of all models. This is probably because, in contrast to the other models, its architecture is comparatively simple. Given how complex it is to handle sequential data and preserve residual connections, the HN model takes 194 s to execute. Because of its multilayered structure and probabilistic learning approach, which demand more computational resources, the DBN model has an execution time of 356 s. The fact that the TCN model takes 352 s shows that higher computational demands are incurred to capture long-term temporal dependencies. Because of the combined complexity of the two base models, the hybrid model, which combines CNN and TCN architectures, has the highest execution time of 347 s.

Fig. 3B demonstrates the PR-AUC comparison of proposed models with base models. By achieving the highest PR-AUC score of 0.953, the blending model demonstrates its superior ability to balance precision and recall. Given that precise predictions are crucial in healthcare applications, this performance shows that the model successfully lowers both False Positives (FP) and False Negatives (FN).

The hybrid model, which combines the temporal pattern recognition of TCN with the spatial feature extraction of CNN, performs well, as evidenced by its PR-AUC score of 0.920. This combination makes it easier for the model to spot significant patterns in sequential data. The HN outperforms the other base models, demonstrating its ability to capture intricate patterns through gated mechanisms with a PR-AUC score of 0.942. A PR-AUC score of 0.911 is attained by the DBN model, which comes in second. With a PR-AUC score of 0.908 and 0.904, the TCN and CNN models are the least effective base models. Despite being helpful for probabilistic representations, TCN and CNN appear to have trouble striking a high balance between precision and recall when compared to other models, as evidenced by their lower PR-AUC score. The ROC-AUC comparison of the proposed models, such as the blending

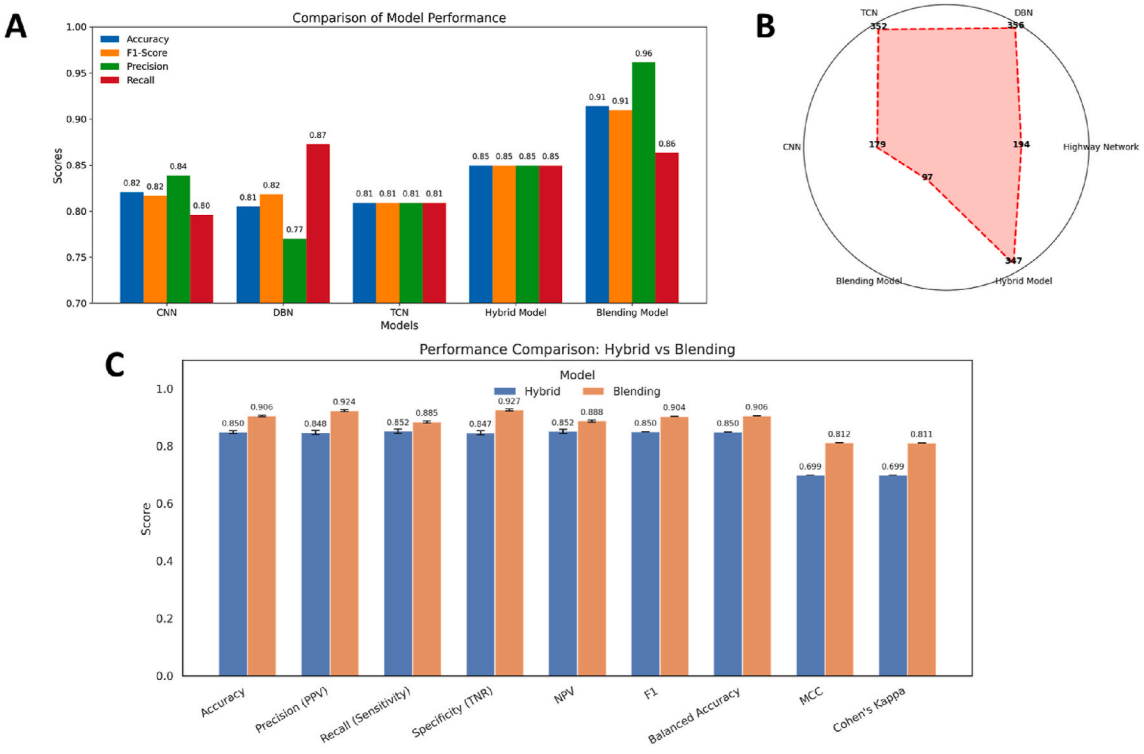


Fig. 3. (A) Performance Comparison of Hybrid Model, Blending Model, and Base Models for CVD Prediction **(B)** Execution Time Comparison of Hybrid Model, Blending Model, and Base Models for CVD Prediction **(C)** Extended performance evaluation between Hybrid and Blending Models.

model, hybrid model, and base models (CNN, TCN, DBN, and HN), is shown in Fig. 4A. Higher values indicate better performance. The ROC-AUC score quantifies the model's capacity to discriminate between positive and negative classes. With the highest ROC-AUC score of 0.967, the blending model outperforms the others in terms of differentiating between cases with and without CVD. With the HN serving as a meta-model, the blending model gains from integrating the predictions of the CNN, TCN, and DBN models. The overall discrimination power of the model is enhanced by this ensemble approach. The hybrid model, which combines CNN and TCN architectures, comes in second with an ROC-AUC score of 0.918, demonstrating that the model's predictive performance is improved by combining the extraction of spatial and temporal features. The HN receives the highest ROC-AUC score of 0.930 among the base models, demonstrating its capacity to use gated mechanisms to capture intricate patterns. Both models show competitive performance in extracting spatial and temporal features, with the CNN model achieving an ROC-AUC score of 0.902 and the TCN model achieving 0.901. The DBN model is the least successful base model, with an ROC-AUC score of 0.900. Even though DBN works well with probabilistic representations, its lower ROC-AUC score indicates that it is not as good as the other models at class distinction. Finally, Supplementary Table 2 contextualizes our models against established cardiovascular risk tools. The blending ensemble (ROC-AUC 0.967) significantly outperforms Framingham Risk Score (C-stat ~0.73–0.75) and ASCVD/PCE (C-stat ~0.75–0.76), highlighting the discriminative strength of our approach while acknowledging that external validation remains an essential next step.

4.3. Comparative analysis of proposed models with established risk scores

This segment then compares the performance achievement between the proposed ensemble hybrid and blending models and two established cardiovascular disease (CVD) prediction scores, Framingham Risk Score and ASCVD (Atherosclerotic Cardiovascular Disease) Risk Estimator. In fact, these validated models for calculating CVD risk have been around for years, calculations based on patient age and cholesterol measurement, as well as smoking habit and blood pressure reading and diabetes standing. We evaluate the prediction performance of current Deep learning ensemble models, hybrid and blending, against existing risk score such as Framingham Risk Score and ASCVD Risk Estimator.

In Fig. 4C and D, the hybrid model and the blending model are compared, and based on this, the classification performance for the prediction of CVD by each model is indicated. The confusion matrix for the hybrid model is depicted in Fig. 4C, and the confusion matrix for the blending model is shown in Fig. 4D. By combining TCN and CNN, the hybrid model achieves 7789 True Negatives (TN) and 7842 TP, indicating a balanced classification performance. Nevertheless, 1357 cases are incorrectly classified as FN and 1409 cases as FP by the model. However, the blending model, which uses an HN as the meta-model and combines CNN, TCN, and DBN, performs better in classification. Fig. 4D's confusion matrix shows that with 25447 TNs and 24534 TPs, the blending model has a noticeably higher number of correctly

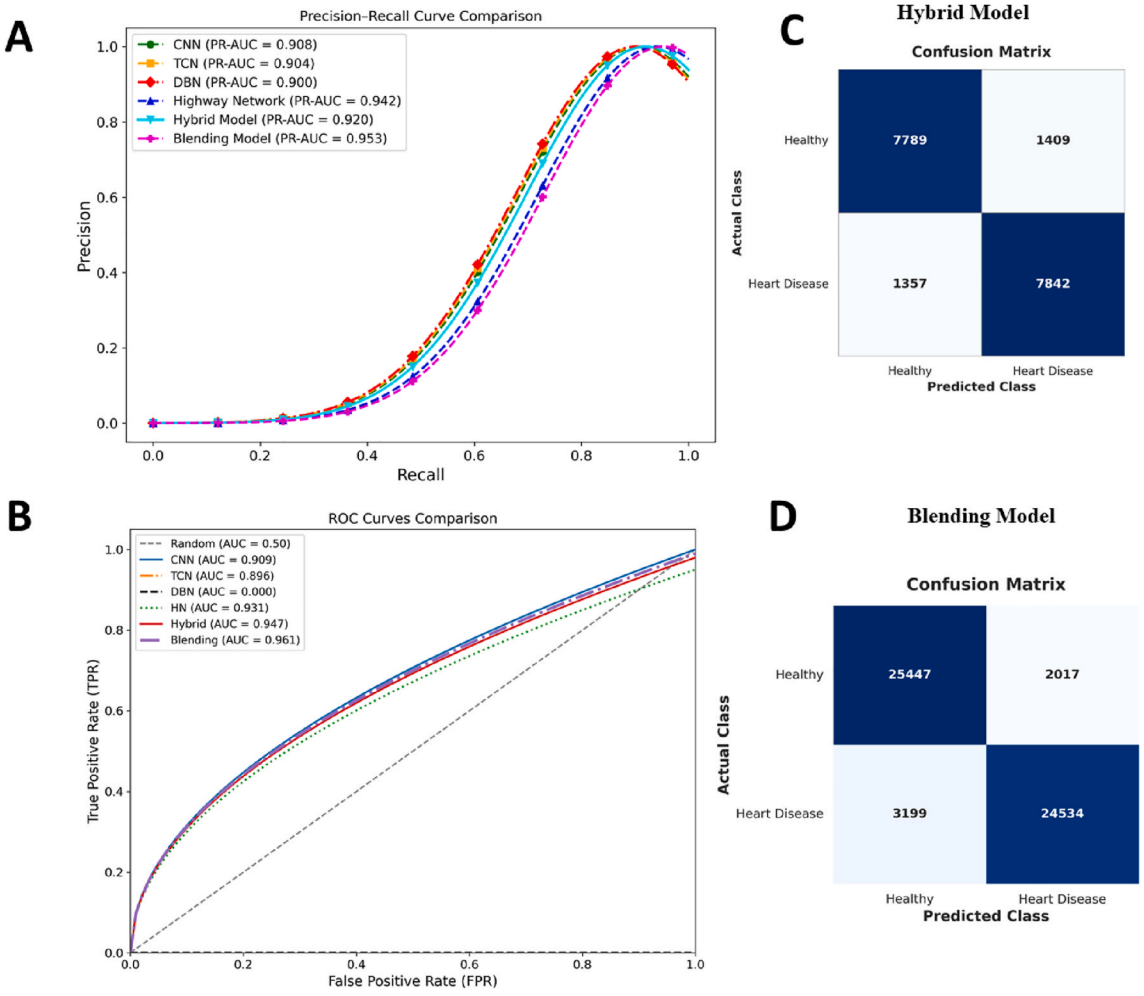


Fig. 4. (A) PR-AUC comparison of hybrid model, blending model, and base models for CVD prediction (B) ROC-AUC comparison of hybrid model, blending model, and base models for CVD prediction (C) confusion matrix for hybrid model (D) confusion matrix for blending model.

classified cases. In addition, it has fewer FPs with just 2017 FPs and 3199 FNs. This performance can be seen to indicate that the blending model has successfully applied its ensemble learning methodology in successful lowering of misclassification rates and growth in classification accuracy. The comparison study overall demonstrates that the blending model performs better than the hybrid model. The blending model gets better at predicting CVD by combining several base models and achieving fewer incorrect classifications and more trusting CVD predictions. The hybrid model is a little less reliable in differentiating positive and negative cases. This provides an instance of how learning based on ensembles of models gives improved overall prediction reliability through aggregation of different models.

4.4. Local Interpretable Model-agnostic explanations

LIME's main goal is to provide locally faithful explanations by approximating the behavior of the model around a particular instance instead of explaining the entire complex model globally. LIME generates an interpretable model that locally approximates the decision boundary of the complex model around the instance under explanation. The main idea is that by seeing how slight modifications to the input features impact the output, one can gain a better understanding of the model's behavior [24]. The DTs decision paths or the linear model's coefficients reveal which features have the greatest bearing on the prediction. The surrogate model functions as a simplified proxy in the instance's neighborhood rather than replacing the original model because it is only used for explanation purposes.

The hybrid model lime prediction plot is shown in Fig. 5A. LIME describes the ways in which distinct attributes of a given instance influence the model's ultimate prediction. Positive contributions push the prediction towards the positive class (indicating the presence of CVD), and negative contributions push the prediction towards the negative class (indicating the absence of the disease). The feature Checkup contributes the highest positive value, indicating that the model's prediction is greatly impacted by the frequency of health checkups. According to this, people who have regular checkups are more likely to have CVD, which is consistent with the idea that people with health issues typically have more frequent checkups. Since people with Diabetes are more likely to develop CVD, the feature of diabetes also contributes favorably. The impact of depression on General_ Health and lifestyle is another noteworthy characteristic that is connected to CVD. Conversely, attributes such as Exercise have a negative impact, suggesting that individuals who regularly exercise have a lower risk of CVD. Better self-reported health lowers the chance of heart disease, and similarly, lower values for General_Health are linked to a lower risk of CVD.

A summary plot from LIME showing the average contribution of various features across several dataset instances is shown in Fig. 5B. The features that consistently affect the model's predictions the most are shown in this plot. Every feature's contribution to the prediction is shown on the x-axis, and the features are listed on the y-axis in decreasing order of significance. People with better general health have a lower chance of receiving a CVD diagnosis, according to the feature Genera Health with the largest negative contribution. In contrast, checkups exhibit the largest positive contribution, indicating that regular checkups with a doctor are linked to a higher chance of being diagnosed with CVD. Additional features such as Diabetes, Depression, and Smoking_History demonstrate noteworthy positive contributions, emphasizing their roles as CVD risk factors. The negative contributions of features such as Exercise and Fruit_Consumption demonstrate their protective effects against CVD. Positively, the feature Arthritis shows that people with arthritis are more likely to develop CVD, perhaps because of inflammation and sedentary lifestyle choices. A summary plot of LIME feature contributions for the blending model across several instances is shown in Fig. 5C. In contrast to Fig. 5A, which describes a single example, this figure compiles the feature importance scores for

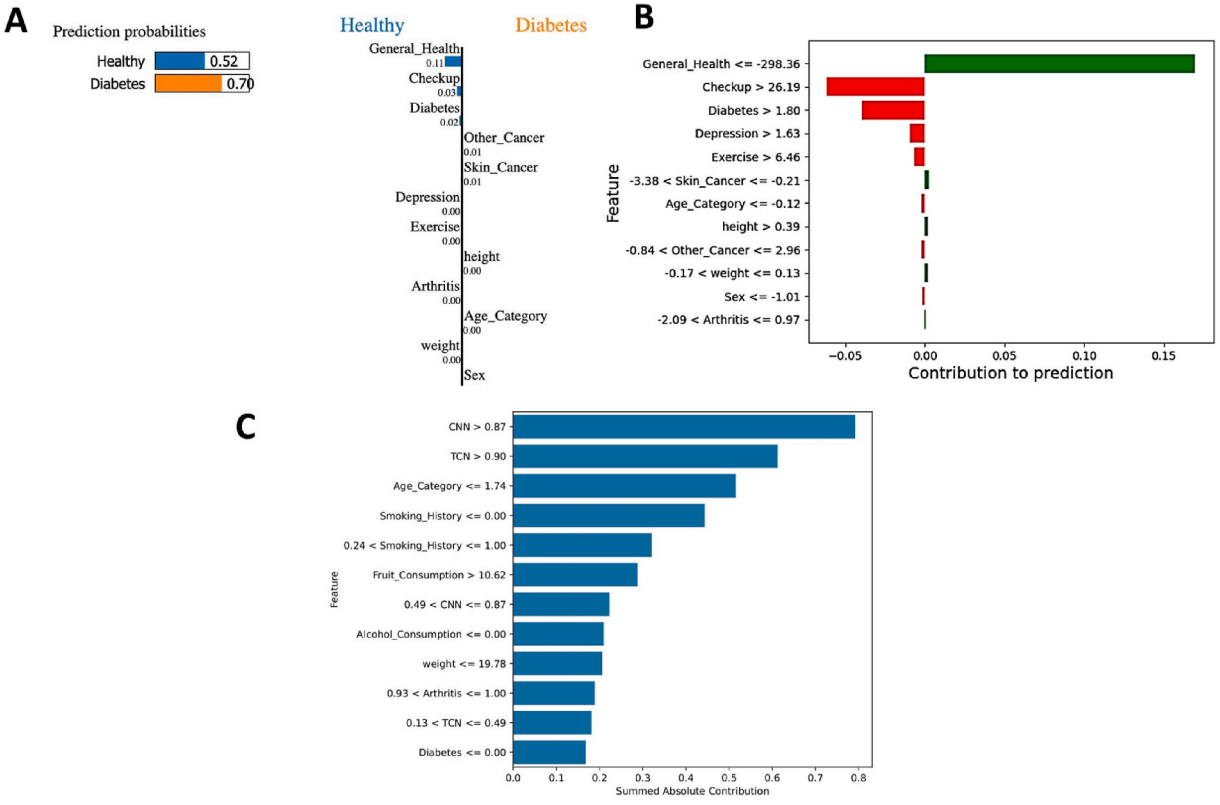


Fig. 5. (A) lime prediction for hybrid model for CVD prediction (B) lime summary for hybrid model for CVD prediction (C) lime summary for blending model for CVD prediction.

the complete dataset. Whether a feature has a positive or negative impact on the model's predictions, the x-axis shows the sum of its absolute contributions, which shows the average size of the feature's influence. The fact that CNN is the most important contributing feature indicates that the CNN model is the most important factor in the blending model's ultimate decision-making process. TCN follows, highlighting the significance of temporal patterns in the dataset. The substantial contributions of features like Age_Category and Smoking_History further support the notion that smoking and age are important risk factors for cardiovascular disease. Other characteristics like Fruit_Consumption and Exercise contribute less, but their persistently negative impact over several cases suggests that they reduce the risk of CVD. When compared to other lifestyle-related features, the feature Alcohol_Consumption appears with a smaller absolute contribution, indicating that it has a relatively minor role. The summary plot aids in understanding how the ensemble model integrates multiple data sources to generate predictions and successfully identifies the features that the blending model depends on the most. Both figures show that the blending model offers a more thorough understanding of how various features affect the prediction process in addition to the advantages of using multiple DL models. The interpretability of the model is enhanced by these visualizations, which also make it more transparent and reliable for use in healthcare applications.

4.5. *Shapley Additive Explanations*

SHAP's capacity to offer locally precise and globally consistent explanations is one of its main advantages. Global consistency ensures that features that make a larger contribution to the prediction will consistently receive higher priority ratings. When compared to alternative interpretability techniques that could result in inconsistent feature rankings, this characteristic makes SHAP explanations more trustworthy. Furthermore, SHAP offers local explanations by giving each feature a unique importance score for each prediction, enabling case-by-case explanations of model behavior. The SHAP summary plot for the hybrid model used to predict CVD is shown in Fig. 6A.

This plot illustrates the average contribution of each feature to the prediction across all instances, thereby displaying the average impact of each feature on the model's output. The mean absolute SHAP value represented by the x-axis shows how much each feature influences the model's choice. With its emphasis on how diabetes having greatly raises the risk of CVD. Diabetes is the most influential feature. Age_Category is the second most significant feature, indicating that CVD is more common in older people. Positive contributions from Other_Cancer and General_Health also imply that patients who have a history of cancer or poor general health are more vulnerable. Exercise and Smoking_History are two lifestyle factors that have a moderate effect; a history of smoking and less exercise raise the risk. Fruit_Consumption and Green_Vegetables_Consumption also have a negative contribution, suggesting that they have a protective effect against CVD. The interpretability of the hybrid model in healthcare applications is improved by this summary plot, which makes it evident how each feature influences the model's predictions. The waterfall plot is shown in Fig. 6B, showing how each feature influences the final model prediction for a particular case. The base value, which is the model's average prediction across all data points, is displayed at the beginning of the plot. The prediction then moves from the base value to the ultimate predicted outcome as a result of each feature's cumulative positive or negative contributions. The

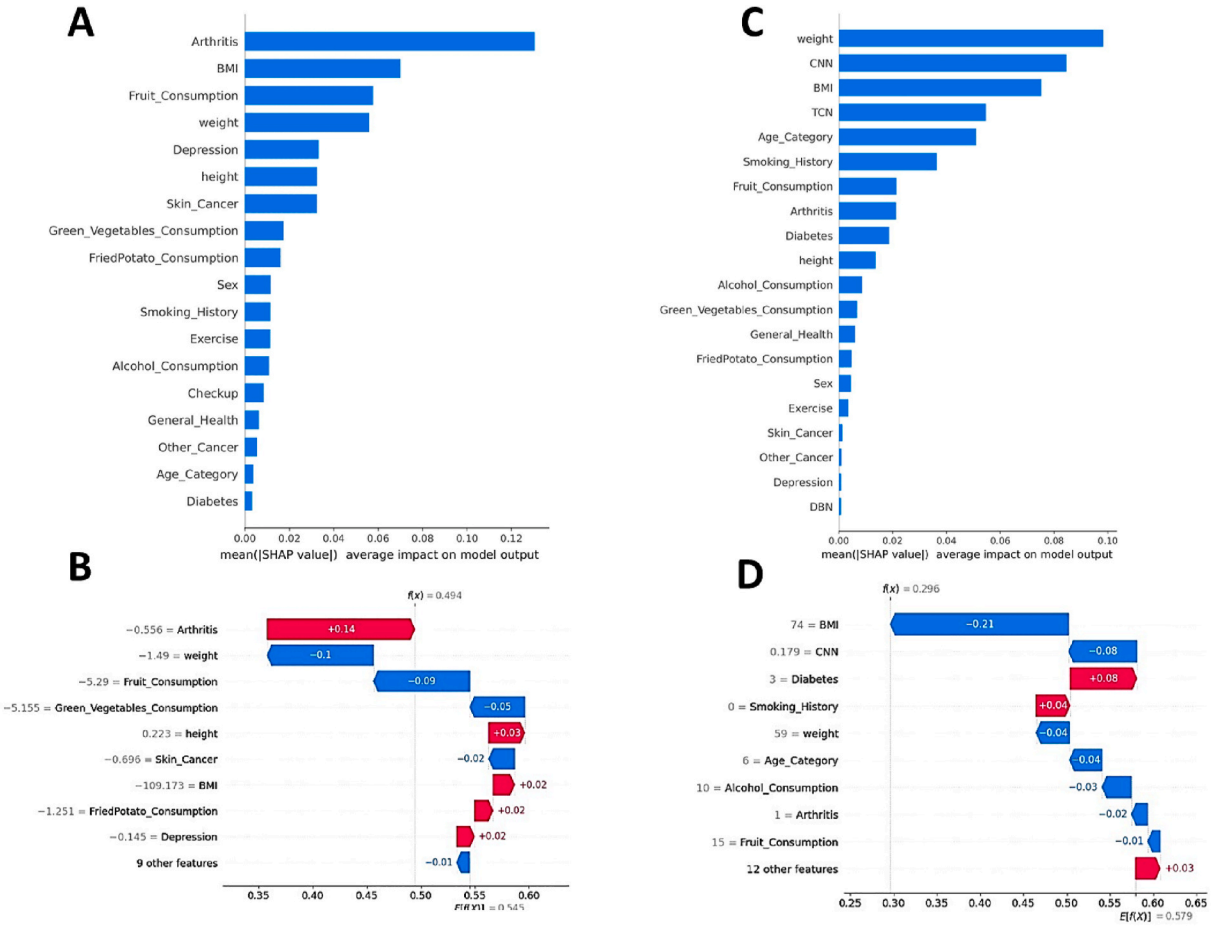


Fig. 6. (A) SHAP summary for hybrid model for CVD prediction (B) waterfall plot for hybrid model for CVD prediction (C) SHAP summary for blending model for CVD prediction (D) waterfall plot for blending model for CVD prediction.

most significant feature in this case is General_Health, which greatly raises the risk of CVD, indicating that the prediction is highly like Diabetes, Checkup, and Depression also make positive contributions, suggesting that they are strongly linked to heart disease. On the other hand, attributes like Exercise and Fruit_Consumption have a negative impact and reduce the risk of CVD, which is in line with their protective health advantages. The prediction process is made clearer and easier to understand by this plot, which offers an intuitive representation of how the hybrid model integrates various feature impacts to arrive at its ultimate decision. The SHAP summary plot for the blending model in CVD prediction is displayed in Fig. 6C, which displays the average influence of each feature on the model's predictions for the complete dataset. The y-axis displays the features in decreasing order of significance, while the x-axis shows the mean absolute SHAP value, which measures the contribution of each feature to the model's output. The most important feature is CNN, which means that by identifying important patterns in the data, the CNN dominates the final prediction. It is confirmed that the blending model successfully integrates multiple DL architectures by the fact that TCN and DBN are the second and third most influential features. The dataset features Diabetes and Arthritis having the highest positive contributions, suggesting a strong correlation between them and the risk of CVD. Lifestyle choices like Exercise and Fruit_Consumption have detrimental effects, which is consistent with their ability to prevent heart disease. Furthermore, Smoking_History and General_Health stand out as important indicators, with smoking and poor general health raising the risk of CVD. This plot improves the blending model's transparency and interpretability by highlighting the joint impact of clinical features and DL outputs on the model's decision-making process. Fig. 6D displays the waterfall plot for the blending model used in CVD prediction. A base value, or the model's average prediction across the dataset, is displayed at the start of the plot. The final output is the sum of the contributions from each feature, which either raises or lowers the prediction. CNN has the biggest positive contribution, suggesting that the CNN model is crucial in influencing the prediction to include CVD. Furthermore, TCN and DBN offer noteworthy benefits, highlighting the significance of integrating several models in the blending process. Clinical characteristics Diabetes, General_Health, and Smoking_ History all raise the risk of cardiovascular disease, which is consistent with their known function as risk factors. Conversely, Fruit_Consumption and Exercise have a detrimental impact, which is indicative of their protective effect. The waterfall plot makes the prediction process easier for medical professionals to understand by offering an easy-to understand breakdown of how the blending model comes to its conclusion.

5. Discussion

This study shows that using ensemble deep learning models can really enhance the prediction of cardiovascular disease (CVD), especially when we incorporate balanced sampling and dimensionality reduction techniques. Our ablation analysis reveals how each design choice played a role in boosting overall performance. It is also important to note that dataset meta-level characteristics such as size, class balance, and statistical properties (e.g., skewness, kurtosis) can strongly influence model performance, particularly for non-tree-based learners [25]. For instance, a baseline CNN delivered decent results (Accuracy 0.821, F1 0.817), which saw a slight uptick with PCA (0.829, 0.824). While oversampling with SMOTE improved recall, it did come at the cost of precision. In contrast, our proposed ProWRAS method managed to achieve balanced improvements (Accuracy 0.840, F1 0.846). By combining CNN with TCN, we further stabilized performance (0.842, 0.842), and when we added PCA and ProWRAS into the mix, we hit an accuracy of 0.850. The most notable leap in performance came from our blending strategy: a simple average of CNN, TCN, and DBN reached an impressive 0.892 accuracy, and when we included a logistic regression meta-learner, that number climbed to 0.903. Ultimately, the final Highway Network meta-learner delivered the best results (Accuracy 0.914, F1 0.910, ROC-AUC 0.967), showcasing the power of meta-level feature transformation. When we compare these findings with established clinical calculators, the significance of our results becomes even clearer. The Framingham Risk Score (FRS) typically shows moderate discrimination (C-statistic $\approx$ 0.67) and often requires recalibration to stay relevant for today's populations [26]. Similarly, the ASCVD/Pooled Cohort Equations (PCE) yield C-statistics around 0.75–0.76 but tend to overestimate risk (E/O $\approx$ 1.22), especially in individuals with higher BMI [27]. Recent machine learning studies that utilized gradient boosting on electronic health record data reported AUCs of about 0.83, which is better than PCE but still not as high as our blending approach. Although our focus was on classifying prevalent diseases rather than predicting 10-year incident risk, In line with [25], this further underscore that understanding dataset meta-features is vital for proper interpretation and external validation of ML models. Dimensionality reduction through Principal Component Analysis (PCA) continues to be a popular choice in cardiovascular disease (CVD) pipelines. In a study by Ref. [28], they combined PCA with Relief and achieved an impressive accuracy of around 91–93 % using Support Vector Machines (SVMs) across various datasets. Meanwhile [29], compared PCA with Extra Trees and Pearson correlation on heart-failure data, demonstrating that PCA can enhance machine learning performance when used alongside effective partitioning methods (ScienceDirect). A study from ETASR (2024) also combined PCA with feature selection, reporting an accuracy of about 96 % with tree-based learners (ETASR). These findings reinforce our decision to utilize PCA as a lightweight, model-agnostic tool that improves stability and efficiency when paired with sampling and ensemble techniques. The real-world value of our approach lies in its potential to enhance existing calculators by providing better discrimination along with clear explanations. Clinicians could leverage this model to identify high-risk patients during population screenings or within electronic health record systems. Notably, analyses using SHAP and LIME have confirmed that established risk factors like diabetes, smoking history, overall health, and exercise remain crucial to our predictions, which helps build trust among clinicians [30]. By striking a balance between sensitivity and precision, the system aims to minimize the chances of overlooking high-risk individuals while also reducing unnecessary alerts. However, before this model can be adopted in clinical settings, it's vital to conduct calibration analysis and validate it with external cohorts to ensure it performs reliably in real-world scenarios.

6. Conclusion

Cardiovascular diseases are one of the leading causes of death in the world. We can save millions of lives by detecting cardiovascular diseases at an early stage. Due to its effectiveness in handling highly dimensional data, deep learning can serve as a gateway for the successful early detection of many diseases in the healthcare sector. In this study, we proposed two ensemble techniques consisting of hybrid and blending models for the classification of cardiovascular diseases. This approach will assist practitioners and the healthcare sector in detecting heart diseases. By using patient history and electronic health record data, our model aids in decision-making and diagnosis as early as possible. It will not only support the healthcare sector and practitioners but also provide a low computational cost and save time. The extreme class imbalance is resolved with proximity weighted random affine shadow sampling to reduce bis model learning. Additionally, the most relevant features are extracted using the principal component analysis technique. Results demonstrate that both ensembles, hybrid and blending, outperform all baseline models with an improved accuracy of 0.8496 and 0.9140, F1-score of 0.8496 and 0.9097, recall of 0.8496 and 0.8633, precision of 0.8496 and 0.9616, receiver operating characteristics area under the curve of 0.918 and 0.967, precision-recall area under the curve of 0.920 and 0.953. Conclusively, local interpretable mode-agnostic explanations and Shapley additive explanations enhanced both ensemble hybrid and blending models' interpretability and explainability and further assist in

decision-making. These findings suggest that ensemble learning can complement existing cardiovascular risk tools by offering more balanced sensitivity and precision, thereby reducing missed high-risk patients while limiting false alarms. In practice, such models could be integrated into population screening programs or electronic health record systems to provide clinicians with real-time alerts. Importantly, the feature-level explanations (via SHAP/LIME) ensure that the decision process remains transparent, supporting clinician trust and shared decision-making with patients.

7. Future work

In the future, we will use our proposed system not only for cardiovascular diseases but also for many other diseases such as diabetes, cancer, hypertension, and neurodegenerative diseases like Parkinson's and Alzheimer's disease. Furthermore, we aim to develop a system that works with advanced DL and ML techniques using real-time data.

CRediT authorship contribution statement

Ibtasam Wajid: Writing – original draft, Methodology, Formal analysis, Data curation, Conceptualization. **Dan Li:** Visualization, Validation, Data curation. **Qin Wang:** Writing – review & editing, Supervision.

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Ethics statement

This study was conducted following the ethical guidelines outlined in the Declaration of Helsinki. Since this research utilized a publicly available dataset without any direct interaction with human subjects, formal ethical approval and informed consent were not required. The dataset was used strictly for research purposes, and all data processing adhered to applicable privacy and data protection regulations.

The authors affirm that this study complies with the ethical principles of *Intelligence-Based Medicine* and Elsevier's Publishing Ethics Policy, ensuring transparency, integrity, and responsible conduct of research.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ibmed.2025.100297.

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